

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, low power usage and easy to use.

4 Port 4 DI + 1 DO Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	12 Pin 5.08 mm pitch pluggable screw Terminal Block, 2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	110 mm L x 70 mm B x 50 mm H
Power	Input Power – 12 - 24 VDC or V 24 AC/DC Typical – 12V DC @ 110mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DI Inputs –

Channels	4
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Relay Output –

Channels	1
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 48 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10 ⁵
Mechanical Life	1 x 10 ⁷
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

Note: -

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Additional Features: -

Communication port isolated
 Input power reverse polarity safety
 ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air
 EFT IEC 61000-4-4, 50A (5/50ms)
 750V isolation.
 CRC Error check.
 No configuration needed on the IO board

Configuration Settings: -

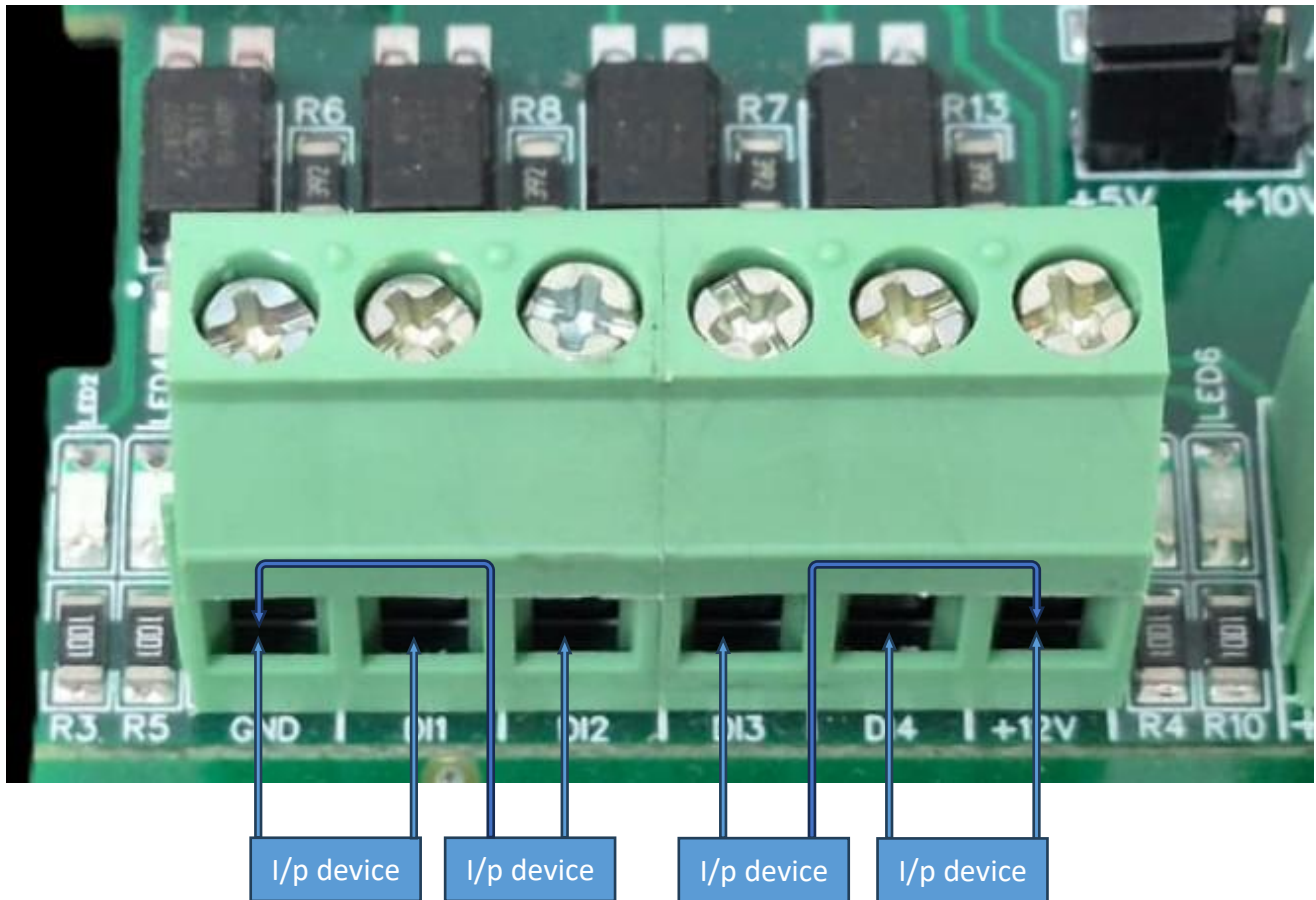
Communication Speed 9600 – 115200 (DIP SW selectable)
Data Bits 8
Parity None
Stop bit 1
CRC Yes
Slave ID 1 – 15 (DIP SW selectable)
Function code DI 0x02 Read discrete Input
DI Register Address 1,2,3,4.
Function code 0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address 0.

Recommended Cable Electrical Characteristics: -

22 AWG Cable Shielded and twisted pair should be used.
Tinned Copper Recommended
Nominal Conductor DCR 14.7 ohm / 1000 ft
Nominal Capacitance 11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss 0.5db / 100ft

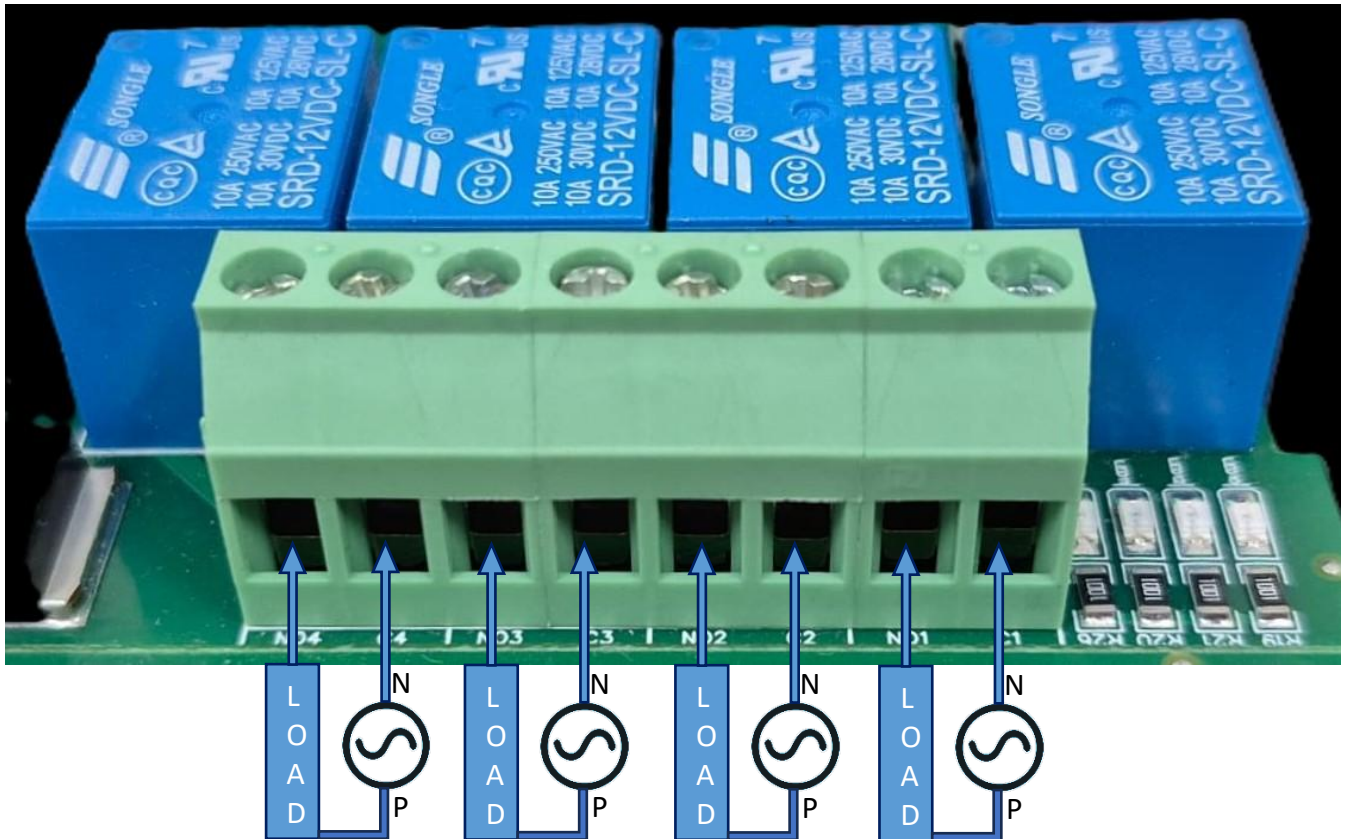
ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	DO 1	00001	0X05,0X0F	RS485	1 Bit Boolean
2	DI 1(With & Without Potential)	10002	0X02	RS485	1 Bit Boolean
3	DI 2(With & Without Potential)	10003	0X02	RS485	1 Bit Boolean
4	DI 3(With & Without Potential)	10004	0X02	RS485	1 Bit Boolean
5	DI 4(With & Without Potential)	10005	0X02	RS485	1 Bit Boolean

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

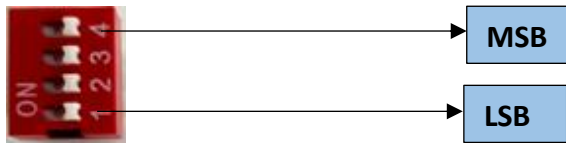
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO** remains open, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The load is connected between terminal **NO** and **Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The load is connected between terminal **NO** and **Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

SLAVE ID DESCRIPTION



For Slave ID Selection SW is used to Set The SLAVE ID .

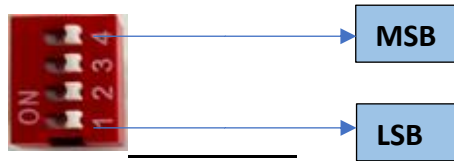
For Slave ID DIP Switch **LSB** is "1" follow through "4" is **MSB**.

Slave ID Confirmed through below Device ID table .

IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So "1 0 0 0" will be selected as Slave ID 1.

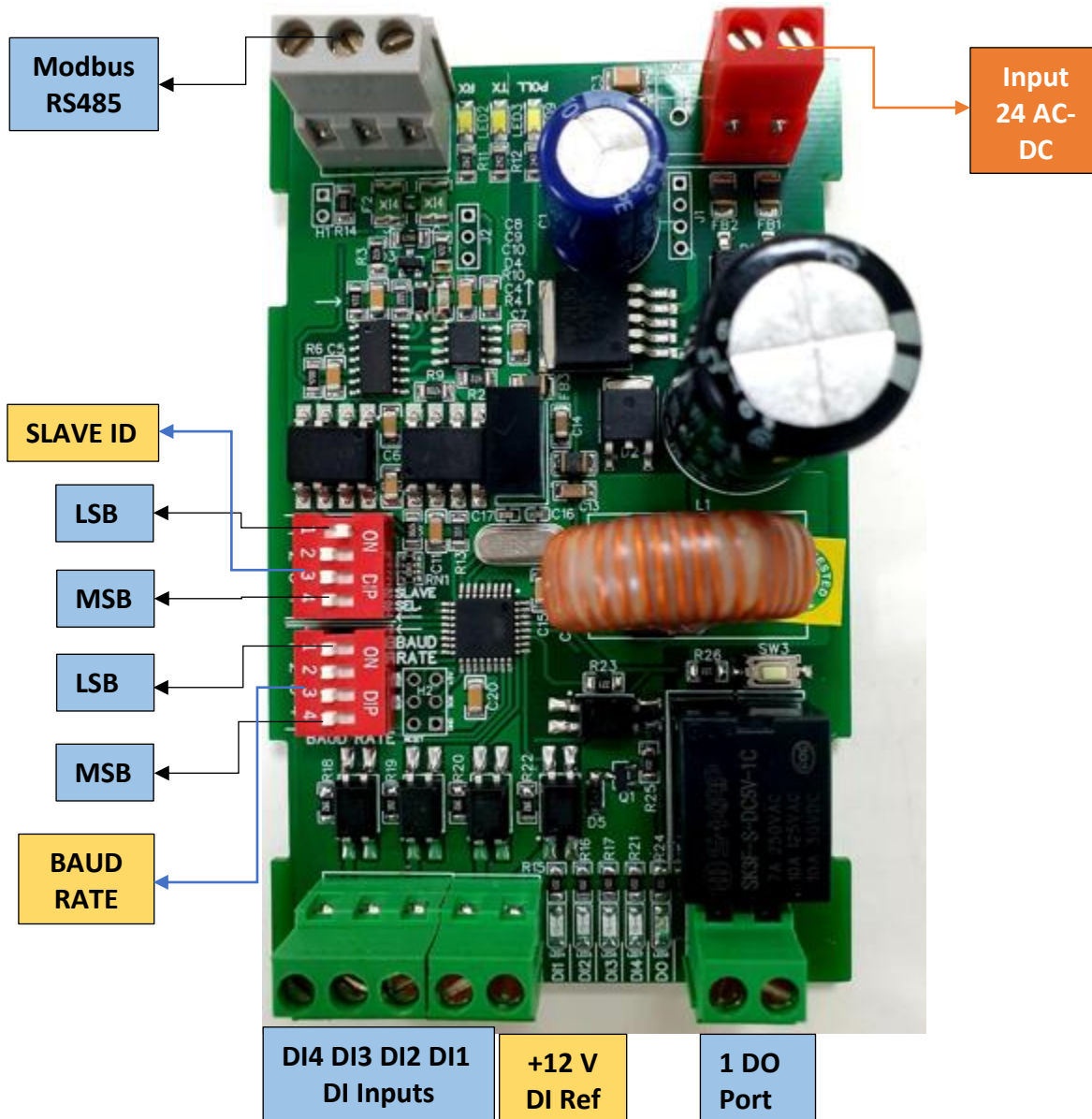
Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 0 1 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	0 0 1 1	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 1 0 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	0 1 0 1	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	0 1 1 1	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	ON(1)	OFF(0)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15

SLAVE ID DESCRIPTION



- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When u change the Baud rate in the Module power 'ON' condition, pls press the reset button to get Change to affect.

Baud Rate	DIP SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
19200	ON	OFF	OFF	OFF
38400	OFF	ON	OFF	OFF
57600	OFF	OFF	ON	OFF
115200	OFF	OFF	OFF	ON



Contact us: -

Augmatic Technologies Pvt. Ltd.,
Plot no 6, Shah Industrial Estate II,
Kotambi,
Vadodara – 391510.

Email – Sales@wittelb.com

Support: support@wittelb.raiseaticket.com / +91 94038 92805